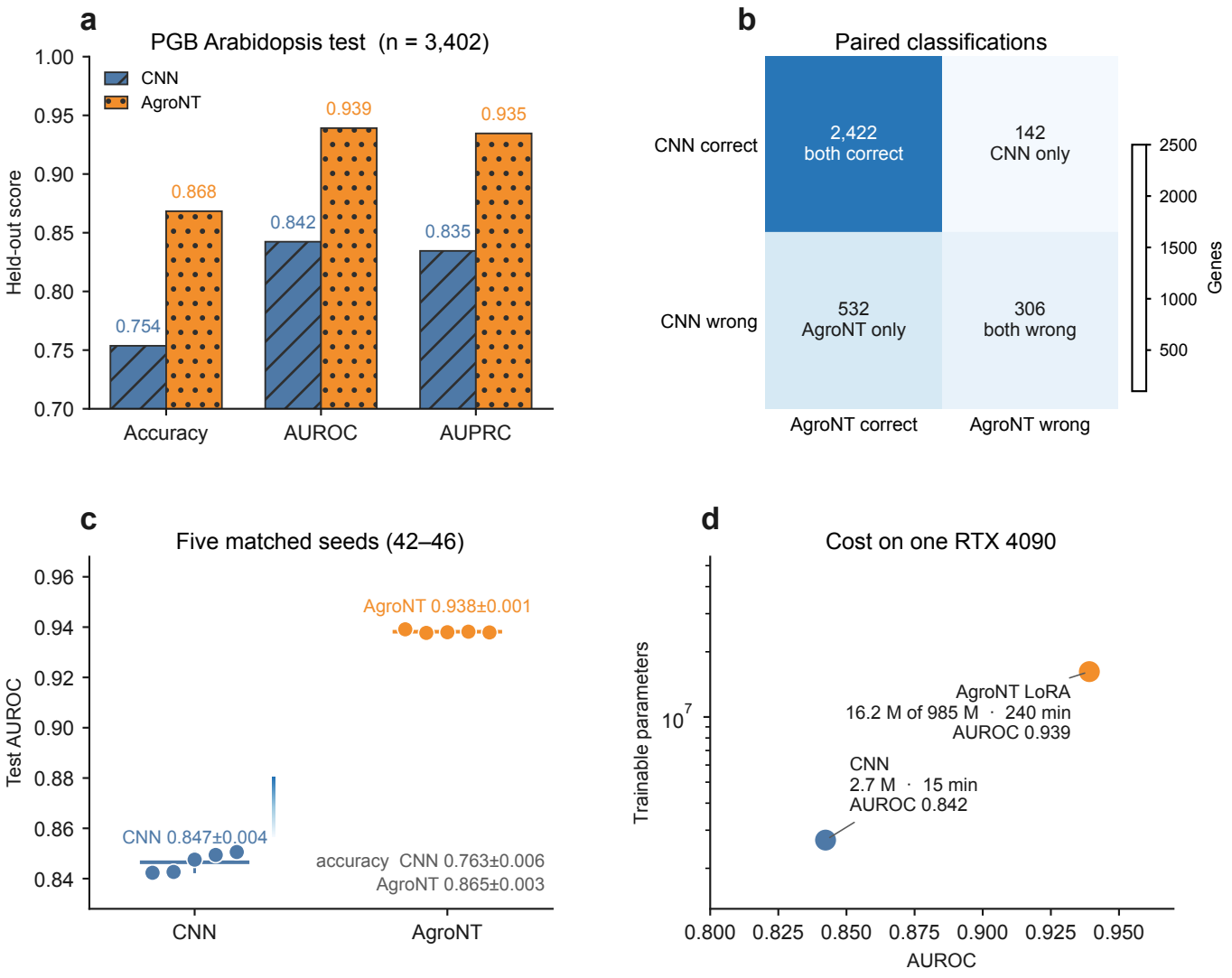
a
PGB Arabidopsis test (n = 3,402)
CNN
AgroNT
Held-out score
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.754
0.868
0.842
0.939
0.835
0.935
Accuracy
AUROC
AUPRC
b
Paired classifications
CNN correct
CNN wrong
2,422
both correct
142
CNN only
532
AgroNT only
306
both wrong
AgroNT correct
AgroNT wrong
2500
2000
1500
1000
500
Genes
c
Five matched seeds (42–46)
Test AUROC
0.96
0.94
0.92
0.90
0.88
0.86
0.84
AgroNT 0.938±0.001
CNN 0.847±0.004
accuracy CNN 0.763±0.006
AgroNT 0.865±0.003
CNN
AgroNT
d
Cost on one RTX 4090
Trainable parameters
10^7
AgroNT LoRA
16.2 M of 985 M · 240 min
AUROC 0.939
CNN
2.7 M · 15 min
AUROC 0.842
0.800
0.825
0.850
0.875
0.900
0.925
0.950
AUROC